LEC03 - Runtime Analysis

Recall

Let $f, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$ be functions

- We say f is bounded above by a positive (constant) multiple of g iff there exists a positive real number C such that

  $f(n) \leq Cg(n)$, for all $n \in \mathbb{N}$

- We say f is eventually bounded above by a positive (constant) multiple of g iff there exists a non-negative integer n_0 such that

  $f(n) \leq g(n)$, for all $n \in \mathbb{N}, n \geq n_0$

Big-Oh

Let $f, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$, we say f is eventually bounded above by a (positive) constant multiple of g, or just $f \in \mathcal{O}(g)$ if there exists a positive real number C and a natural number n_0 such that

$f(n) \leq Cg(n)$ for all $n \geq n_0$, $n \in \mathbb{N}$

Sometimes, we denote it by $f \preceq g$

Recall

Let $f, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$ be functions

- We say f is bounded below by a positive constant multiple of g iff

  $f(n) \geq Cg(n)$, for all $n \in \mathbb{N}$

- We say f is eventually bounded below by g iff there exists a natural number n_0 such that

  $f(n) \geq g(n)$, for all $n \in \mathbb{N}, n \geq n_0$

Big Omega

Let $f, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$, we say f is eventually bounded below by a positive constant multiple of g, or just $f \in \Omega(g)$ if there exists a positive real number C and a non-negative integer n_0 such that

$f(n) \geq Cg(n)$ for all $n \in \mathbb{N}, n \geq n_0$

Sometimes we denote it by $f \succeq g$

Big Theta

Let $f, g : \mathbb{N} \to \mathbb{Z}_{\geq 0}$, we say f is eventually bounded above and below by positive constant multiples of g, or just $f \in \Theta(g)$ if there exists positive real numbers C, C' and natural numbers n_0, n_0' such that

$f(n) \leq Cg(n)$, for all $n \in \mathbb{N}, n \geq n_0$

$f(n) \geq C'g(n)$, for all $n \in \mathbb{N}, n \geq n_0'$

Example

$$\frac{n(n+1)}{2} = \frac{n^2+n}{2} \in \Theta(n^2)$$

① $\frac{n^2+n}{2} \in O(n^2)$

$$\frac{n^2+n}{2} \le Cn^2$$

Take $C=1$, $n_0=0$

$\frac{n^2}{2} \le \frac{n^2}{2}$ and $\frac{n}{2} \le \frac{n^2}{2}$

$$\frac{n^2}{2} + \frac{n}{2} \le \frac{n^2}{2} + \frac{n^2}{2}$$

$$\frac{n^2+n}{2} \le n^2$$

so $f(n) \in O(n^2)$

② $\frac{n^2+n}{2} \in \Omega(n^2)$

$$Cn^2 \le \frac{n^2+n}{2}$$

take $C = \frac{1}{2}$, $n_0 = 0$

$$\frac{n^2}{2} \le \frac{n^2}{2}$$

$$\frac{n^2}{2} \le \frac{n^2}{2} + \frac{n}{2}$$

$$\frac{n^2}{2} \le \frac{n^2+n}{2}$$

so $f(n) \in \Omega(n^2)$

so $f(n) \in \Theta(n^2)$

Some Properties

Let $\mathcal{F} \stackrel{def}{=} \{f : \mathbb{N} \to \mathbb{R}_{\ge 0}\}$, and let $f, g \in \mathcal{F}$

1. The relation $f \preceq g$ is a partial order. It is the reverse of $f \succeq g$. That is to say, $f \in O(g)$ iff $g \in \Omega(f)$
2. The relation $f \asymp g$ is an equivalence relation, defined by the equivalence classes $O(f)$ for $f \in \mathcal{F}$
3. If $f_1 \in O(g_1)$, and $f_2 \in O(g_2)$, then $f_1 + f_2 \in O(\max(g_1, g_2))$
4. If m is any integer with $m > 0$ and $a_0, a_1, a_2, \cdots, a_m$ are real numbers with $a_m \ne 0$, then

$$f(n) = a_m n^m + a_{m-1} n^{n-1} + \cdots + a_1 n + a_0$$

is in $O(n^m)$

Useful Theorem

Let $f, g \in \mathcal{F}$, if

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \begin{cases} C \in \mathbb{R}_{\ge 0} & \text{then } f \asymp g \\ 0 & \text{then } f \preceq g \text{, but } g \not\preceq f \\ \infty & \text{then } f \not\preceq g \text{, but } g \preceq f \end{cases}$$

ex.

$f(n) = \sqrt{n}$, $g(n) = \ln(n)$

$$\lim_{n\to\infty} \frac{\sqrt{n}}{\ln(n)} \stackrel{LH}{=} \lim_{n\to\infty} \frac{\frac{1}{2\sqrt{n}}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{\sqrt{n}}{2} = \infty$$

so $g(n) \preceq f(n)$, so $\ln(n) \in O(\sqrt{n})$

Cases

Let X denote the space of all possible inputs of an algorithm A, and X_n denote the set of all inputs of size n. Let $p(A;x)$ be the number of primitive steps done by the algorithm A for the input x

Best case: $t_*(A;n) \overset{def}{=} \min_{x \in X_n} p(A;x)$

Worst case: $t(A;n) \overset{def}{=} \max_{x \in X_n} p(A;x)$

Average case: $\bar{t}(A;n) \overset{def}{=} \frac{1}{\# X_n} \sum_{x \in X_n} p(A;x)$

Example

Problem: Given a list L of integers, compute the maximum $\max(L)$

```
m ← x_0
for k = 1 to n do
    if x_k > m then
        m ← x_k
    end
end
return m
```

Best case: 1 write

Worst case: n writes

Average case: Let t_{n+1} be the average number of replacements done for list L, by COMPUTE-MAX, then

$$t_{n+1} = \begin{cases} 0 & \text{if } n=0 \\ \frac{1}{2} & \text{if } n=1 \\ t_n + \frac{1}{n+1} & \text{if } n>1 \end{cases}$$

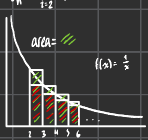
ex. $t_3 = t_2 + \frac{1}{3} = \frac{1}{2} + \frac{1}{3}$

$$t_n = \sum_{i=2}^{n} \frac{1}{i}$$

Problem: How big is t_n for large values of n

Proposition: $t_n \in \Theta(\log n)$

$$t_n = \sum_{i=2}^{n} \frac{1}{i}$$



$\int_2^n \frac{1}{x}\,dx \leq \text{area}$ riemann sum

$\ln(n) - \ln(2) \leq \sum_{i=2}^{n-1} \frac{1}{i}$

$\ln(n+1) - \ln(2) \leq \sum_{i=2}^{n} \frac{1}{i} = t_n$

$t_n \in \Omega(\ln(n))$

$\text{area}(\,) \leq \int_2^n \frac{1}{x}\,dx$

$\frac{1}{3} + \cdots + \frac{1}{n} \leq \ln(n) - \ln(2)$

$t_n - \frac{1}{2} \leq \ln(n) + \ln(2)$

$t_n \in O(\ln(n))$

$t_n \in \Theta(\ln(n))$